MUTHU BALAMURUGAN

AI & Data Science Student | Full-Stack & Generative AI Developer
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SUMMARY

AI & Data Science undergraduate passionate about building high-performance web applications, generative AI systems, and hardware integrations. Experienced in React, Next.js, Python, TypeScript, and intelligent agent workflows.

EDUCATION

B.Tech in Artificial Intelligence & Data Science

Cumulative GPA: 7.8 / 10.0 | Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems, IoT & Embedded Systems

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, HTML5, CSS3
Frameworks & Web: React.js, Next.js, Node.js, Express, FastAPI, Tailwind CSS, REST APIs
AI & Machine Learning: PyTorch, TensorFlow, OpenCV, MediaPipe, LangChain, Hugging Face, Gemini API, Scikit-Learn
Databases & Tools: PostgreSQL, Supabase, MongoDB, MySQL, Git, GitHub, Docker, Postman, Linux
Hardware & IoT: Arduino, ESP32, MQTT, Sensors & Actuators, Embedded C++

FEATURED PROJECTS

EchoVerse AI - Multiverse & Temporal Exploration Platform

- Engineered a multi-agent generative storytelling platform simulating past, parallel reality, and future timelines.
- Integrated streaming LLM APIs with real-time responsive visual graph state rendering in React and FastAPI.

AI Student OS - Unified Academic Productivity Platform

- Architected a comprehensive productivity suite consolidating notes summarization, course roadmaps, and tracking.
- Built real-time sync with Supabase and automated spaced repetition flashcard generation.

Motor Health Monitor - IoT Industrial Anomaly Detection

- Designed an edge IoT predictive diagnostic system with Arduino, vibration sensors, and thermal monitoring.
- Streamed sensor telemetry over MQTT to a Python analytics dashboard with automated threshold alerting.

CERTIFICATIONS & ACHIEVEMENTS

- Deep Learning & Generative AI Specialization - DeepLearning.AI / Coursera (2025)
- Full-Stack Web Development Professional Certificate - Meta / freeCodeCamp (2024)
- Embedded Systems & IoT Engineering - Infosys Springboard (2024)
- 2x Hackathon Finalist & Top Contributor in Collegiate AI Challenges